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Modern statistical and philosophical framework for uncertainty assessment in biometric performance testing

James L. Wayman¹, Antonio Possolo², Anthony J. Mansfield³

¹Office of Graduate Studies and Research, San Jose State University, San Jose, California, USA

²Statistical Engineering Division, National Institute of Standards and Technology, Gaithersburg, Maryland, USA

³Centre for Mathematics and Scientific Computing, National Physical Laboratory, Teddington, Middlesex UK

E-mail: jlwayman@aol.com

Abstract: The question of estimating uncertainty in measurement is fundamental to all scientific fields. In the field of automated human recognition, lack of repeatability and reproducibility of measurements has been noted since at least the 1970s. This study discusses current approaches to estimation of measurement uncertainty within the broader context of scientific philosophy and measurement science. The authors discuss the Duhem–Quine thesis on testing holism and international standards on estimating and reporting uncertainty in laboratory measurements, then apply these concepts to the estimation of uncertainty in technology, scenario and operational testing in biometrics. The authors advocate for moving beyond the calculation of ‘coverage’ intervals as defined in the ISO/IEC ‘guidelines for the expression of uncertainty in measurement’ to full application of the concepts of uncertainty assessment.

1 Introduction

In this paper, our central focus will be on ‘technical testing’ in biometrics – the reporting of recognition mistakes (in the sense of an incorrect decision – either positive or negative – as to whether a biometric sample is from the same characteristic of the same person as the biometric reference). [We accept for the purposes of this paper the definition of biometrics from ISO/IEC JTC1 2382-37:2012 as the ‘automated recognition of individuals based on their behavioural and biological characteristics’, but recognise that the general scientific community defines the term more broadly.] This has been the dominant biometric testing paradigm since 1970. We fully acknowledge, however, that other forms of testing, such as standards-conformance, usability and reliability/availability/maintainability, are closely intertwined with technical testing and may, from an applications point of view, be ultimately more important.

This paper is concerned with the measurement of technical performance of biometric systems – what we measure and report and the uncertainty inherent in these measurements, particularly with regard to predictive value: their repeatability [‘Measurement precision under a set of repeatability conditions of measurement’, meaning ‘a set of conditions that includes the same measurement procedure, same operators, same measuring system, same operating conditions and same location, and replicate measurements on the same or similar objects over a short period of time’ [1].] in future performance of the same system or their reproducibility [‘Measurement precision under

reproducibility conditions of measurement’, meaning ‘conditions that includes different locations, operators, measuring systems and replicate measurements on the same or similar objects’ [1].] in another system. Specifically, we address three questions:

1. Does our measurement reflect the true value of what we wanted to measure?
2. If we repeat the test on the same system, how closely can we expect the same measurement to be obtained?
3. If we reproduce the test on a different system, how closely can we expect the same measurement to be obtained?

We want to bring to this discussion the rich history of assessment and reporting of measurement uncertainty developed by the international statistical and metrological communities.

The organisation of this paper is as follows. In Section 1, we discuss the philosophical underpinnings of the modern interpretation of scientific testing. In Section 2 we discuss the modern development of uncertainty in relationship to statistical control. In Sections 3 and 4, we discuss how the measurement science (metrology) community has come to understand uncertainty and the current international guidelines for the expression of uncertainty in laboratory measurements. Finally, in Sections 5–9, we suggest ways that the international guidelines for uncertainty assessment can be applied to the reporting of test results in biometrics, followed by specific discussion of uncertainty in technology, scenario and operational tests.

2 Duhem–Quine thesis and the theory of holism in scientific testing

If ‘biometrics’ in our sense of the term is to be undertaken as science, then the biometrics research community should follow the historical and philosophical understandings of scientific testing in general. Although much of the history of science has been concerned with the methodology of scientific testing [2–4] and more recently even its anti-methodology [5], we want to focus our discussion on philosophical and statistical underpinnings of testing and reporting metrics. This paper is concerned with measurement and reporting, and how reported measurements should be understood as applying to future performance of the same or different systems [We have learned from our own mistakes in documenting past tests [6–8] that popular misunderstanding of the results seems to be the norm rather than the exception.].

In 1906, the French physicist and philosopher, Pierre Duhem wrote: ‘An experiment in physics can never condemn an isolated hypothesis, only a whole theoretical group’ [9]. Duhem’s point here and throughout his work is that every experiment takes place within a context of theory, apparatus and experimental conditions.

In sum, the physicist can never subject an isolated hypothesis to experimental test, but only a whole group of hypotheses; when the experiment is in disagreement with his predictions, what he learns is that at least one of the hypotheses constituting this group is unacceptable and ought to be modified; but the experiment does not designate which one should be changed [9].

Duhem was writing about physics and explicitly excluding other sciences such as physiology and chemistry, which he considered as less advanced and ‘still close to their origins’. Writing nearly a half century later, without knowledge of Duhem’s work and from a completely different perspective, Quine wrote

The unit of empirical significance is the whole of science....The totality of our so-called knowledge or beliefs...is a man-made fabric which impinges on experience only along the edges. Or, to change the figure, total science is like a field of force whose boundary conditions are experience. A conflict of experience along the periphery occasions readjustment in the interior of the field. Truth values have to be redistributed over some of our statements...But the total field is so underdetermined by its boundary conditions, experience, that there is much latitude of choice as to what statements to reevaluate in the light of any single contrary experience. No particular experiences are linked with any particular statements in the interior of the field except indirectly through considerations of equilibrium affecting the field as a whole [10].

Quine’s paper is considered one of the most influential papers on the philosophy of science in the 20th Century – ‘a modern classic’ [11]. A number of more recent philosophers of science [12] have taken the Duhem and Quine theses together to create a ‘Theory of Holism’ in scientific testing. That theory says that the results of any scientific test reflect the totality of conditions of the test (‘the unit of empirical significance’), including

instrumentation, background assumptions, auxiliary hypotheses and even the theories being tested themselves. So what is measured in any experiment is the totality of all the elements existing in both the physical and intellectual environment of the test. Further, the measurement results must be expressed using words and concepts which themselves may be subject to change as our understanding progresses.

So a ‘technical performance test’ in biometrics measures everything involved in the test – both tangible and intangible. This includes the collection, analysis and computational equipment, the sensors and algorithms, the human subjects including their attitudes/behaviours/beliefs, the physical environment of the test including acoustic noise levels/humidity/day of the week/season/year, all of which are coloured by the underlying theories of what it is that biometric systems do.

3 ‘Statistical control’ in scientific measurement

So if the results of a scientific experiment involving measurement express not only the object of measurement (‘measurand’) but also the effects of all experimental factors, how can an experiment reveal anything useful at all? Addressing this question fully would take us into the realms of the philosophy of science and epistemology in general – we will not pursue such lofty goal [13, 14]. One component of the answer, however, involves the concept of ‘statistical control’, as pioneered by Shewhart [15] and applied by the statistician Eisenhart, who wrote ‘a measurement operation must have attained what is known in industrial quality control language as a state of statistical control... before it can be regarded in any logical sense as measuring anything at all’ [16]. [Eisenhart founded the Statistical Engineering Laboratory of the National Bureau of Standards (NBS) in 1947. In 1988 NBS was renamed as The National Institute of Standards and Technology (NIST).]

For Eisenhart, a central requirement for any scientific measurement was that it be repeatable and reproducible. In calling on the language of industrial quality control, Eisenhart was appealing to the concept that variation can be categorised as stemming from, in the parlance of the time, ‘common causes’ (systematic) and ‘chance causes’ (random). Both types of variation limit the repeatability and reproducibility of scientific measurements. The goal of the experimenter then is to understand and control those causes. This is a state of ‘statistical control’. The resulting measurements require an associated statement of uncertainty stemming from both chance causes and common causes, ‘and their relative importance in relation to the intended use of the reported value, as well as to other possible uses to which it may be put’ [16].

In the context of Duhem–Quine, this means that the tangible factors from a test’s ‘unit of empirical significance’ affecting the observed values of the measurement must be known, controlled and reported. Those that cannot be known, controlled or reported, become part of the uncertainty in the repeatability or reproducibility of the measurements. Without a statement regarding the conditions pertaining in the ‘unit of empirical significance’ being measured, the measurement represents ‘nothing at all’ in the sense of learning anything generalisable beyond the immediate experiment. To learn anything useful from a test,

a state of 'statistical control' within that 'unit' must be attained.

4 International approach to uncertainty in laboratory measurement

So if the unit of empirical significance is the whole of science (or at least some less expansive 'web of belief', as described in a very accessible work created for Quine's undergraduate philosophy courses: Quine WV, Ullian JS. *The Web of Belief*. New York: McGraw-Hill; 1978), then how can the researcher practically attain a state of 'statistical control' over everything all at once so as to avoid measuring nothing at all? As all of the tangible factors influencing the measurements cannot even be listed, 'statistical control' over all of them in any set of experiments cannot possibly be attained. If these factors cannot even be listed, experiments seem relegated to measuring 'nothing at all', unless a way can be found to acknowledge and account for potentially vast areas of unknown and uncontrolled influences creating uncertainty in our measurements.

In the 1960s, several groups internationally grappled with this perplexing state of affairs, with influential works by Ku [17, 18] and Eisenhart [19]. In 1977, the working group on the statement of uncertainties was convened by the Bureau International des Poids et Mesures (BIPM) in response to a request of the Comité International des Poids et Mesures (CIPM). In 1980, that group issued Recommendation INC-1 [20] [Recommendation INC-1, which is a 1980 document, is included as Appendix A in [20], which is a 2008 document.], which was subsequently approved by the CIPM.

The uncertainty in the result of a measurement generally consists of several components which may be grouped into two categories according to the way in which their numerical value is estimated:

- A. those which are evaluated by statistical methods,
- B. those which are evaluated by other means.

There is not always a simple correspondence between the classification into categories A or B and the previously used classification into 'random' and 'systematic' uncertainties. The term 'systematic uncertainty' can be misleading and should be avoided.

Any detailed report of the uncertainty should consist of a complete list of the components, specifying for each the method used to obtain its numerical value [20].

The classification of uncertainty into Types A and B applies to the methods of evaluation, not to the sources of uncertainty themselves. Evaluation by statistical methods means assessment of a component of uncertainty by applying statistics to replicated indications obtained during measurement. Other means of evaluation include information derived from authoritative publications, for example, in the certificate of a certified reference material, or based on expert opinion. In general, Recommendation INC-1 maintained a conventional approach to overall reporting of uncertainty, stating

...The components in category A are characterised by the estimated variances... (or the estimated 'standard deviations'...) and the number of degrees of freedom.... The components in category B should be characterised by quantities ... which may be

considered as approximations to the corresponding variances, the existence of which is assumed. ... The combined uncertainty should be characterised by the numerical value obtained by applying the usual method for the combination of variances. The combined uncertainty and its components should...be expressed in the form of 'standard deviations' [20].

In response to CIPM recommendations, additional papers [21], laid the groundwork for the 1994 ISO Guide to the Expression of Uncertainty in Measurement [22] and the NIST guidelines to evaluate and express measurement uncertainty [23].

5 ISO/IEC Guide 98

The ISO/IEC 'Guide to Expression of Uncertainty in Measurement' (commonly referred to as the GUM) was motivated by the need to establish a single, internationally accepted approach to reporting uncertainty in laboratory measurements. The GUM sets the framework for uncertainty assessment as follows:

'The concept of *uncertainty* as a quantifiable attribute is relatively new in the history of measurement, although *error* and *error analysis* have long been a part of the practice of measurement science or metrology. It is now widely recognised that, when all of the known or suspected components of error have been evaluated and the appropriate corrections have been applied, there still remains an uncertainty about the correctness of the stated result, that is, a doubt about how well the result of the measurement represents the value of the quantity being measured.' [22]

The GUM uses the term 'measurand' to denote the quantity being measured, explaining this as

The first step in making a measurement is to specify the measurand – the quantity to be measured... in principle, a measurand cannot be *completely* described without an infinite amount of information. Thus, to the extent that it leaves room for interpretation, incomplete definition of the measurand introduces into the uncertainty of the result of a measurement a component of uncertainty that may or may not be significant relative to the accuracy required of the measurement.

Commonly, the definition of a measurand specifies certain physical states and conditions.

EXAMPLE The velocity of sound in dry air of composition (mole fraction) $N_2 = 0.780\ 8$, $O_2 = 0.209\ 5$, $Ar = 0.009\ 35$ and $CO_2 = 0.000\ 35$ at the temperature $T = 273.15\ K$ and pressure $p = 101\ 325\ Pa$. [22].

The GUM discusses the measurement of a measurand observed through a test as the 'realised quantity' [A common example from biometrics would be to use the comparison of two images from different sources as a proxy for a 'zero effort impostor attempt'. The observed value of false match rate (FMR) for the non-mated comparisons would be an approximation of the desired measurand, the 'zero effort impostor rate'.]

Ideally, the quantity realised for measurement would be fully consistent with the definition of the measurand. Often, however, such a quantity cannot be realised and the measurement is performed on a quantity that is an approximation of the measurand [22].

In this context ‘uncertainty’ is to be understood generically in its natural language meaning of incomplete knowledge (of the measurand we know only as much as indications obtained during measurement reveal and add to any preexisting knowledge, considering that these indications are themselves affected by uncertainty). This does not suggest the presence of ‘error’ in the sense of ‘mistake’. More specifically, the ‘International Vocabulary of Metrology – Basic and general concepts and associated terms’ [1] defines ‘measurement uncertainty’ as a non-negative parameter characterising the dispersion of the quantity values being attributed to a measurand, based on the information used. This obviously fails to cover cases where the measurand is not a scalar, but can be generalised to address those as well.

The GUM summarises several possible sources of uncertainty in measurement, which affect to varying degrees all forms of biometric testing:

- (a) incomplete definition of the measurand;
- (b) imperfect realisation of the definition of the measurand;
- (c) non-representative sampling – the sample measured may not represent the defined measurand;
- (d) inadequate knowledge of the effects of environmental conditions on the measurement or imperfect measurement of environmental conditions;
- (e) personal bias in reading analogue instruments;
- (f) finite instrument resolution or discrimination threshold;
- (g) inexact values of measurement standards and reference materials;
- (h) inexact values of constants and other parameters obtained from external sources and used in the data-reduction algorithm;
- (i) approximations and assumptions incorporated in the measurement method and procedure; and
- (j) variations in repeated observations of the measurand under apparently identical conditions.

This listing from the GUM encapsulates the primary points being made in this paper, which will be applied in the next section to measurement in biometrics.

1. ‘Uncertainty’ is a broader concept than ‘error’, as historically understood. It is the doubt (reflecting incompleteness of knowledge) about how well the test result represents the quantity measured (or being said to be measured). Uncertainty can exist even in the absence of error in the sense of ‘mistake’ and extends beyond the analytic calculation of expected random variation caused by limited test size.
2. A central source of uncertainty is definitional incompleteness in specifying the ‘unit of empirical significance’ for the measurand – full specification of which would require a ‘infinite amount of information’.
3. What is measured is often only a proxy for the measurand of real interest, even if fully defined, adding yet another source of uncertainty in our measurement.

However, the GUM also says

In practice, the required specification or definition of the measurand is dictated by the required accuracy of measurement. The measurand should be defined with sufficient completeness with respect to the required accuracy so that for all practical purposes associated with the measurement its value is unique [22].

This echoes Eisenhart [19] in stating that the required level of definition and specification of the measurand is limited by the measurement accuracy required by the intended use of the data. If there is a known target application, we may be able to estimate the degree of statistical control required so that experimental measurements will be close enough for the intended use. Hence, the required level of statistical control over the tangible elements in the unit of empirical significance is limited by the uses intended for the data. To the three points above we can now add a fourth:

4. How we control, measure and report the values in a test must reflect how we expect those values to be used by others. In other words, testing and reporting must take into account, and state, how the results are expected to be used.

One of the GUM’s most consequential prescriptions is the suggestion that all assessments of uncertainty components should be treated alike, irrespective of whether they will have been evaluated by Type A or Type B methods. The components are commonly treated as if they were variances or standard deviations, with their natural extensions when the measurand is not a scalar.

This approach gives a highly flexible method for assessing all uncertainty components and using the resulting assessments. In particular, the GUM recognises the possibility of ‘insight based on experience and general knowledge’ producing valid assessments, and notes ‘that a Type B evaluation of standard uncertainty can be as reliable as a Type A evaluation, especially in a measurement situation where a Type A evaluation is based on a comparatively small number of statistically independent observations.’ [22].

Estimates of the contributions from all sources of uncertainty, irrespective of their intrinsic nature and of the method that will have been employed to assess them, must be combined into a ‘combined uncertainty’ taking the form of a standard deviation, duly taking into account any correlations between those sources. The resulting combined uncertainty may then be used to produce an interval that, with stated ‘level of confidence’, is supposed to include the measurand: the endpoints of this interval may be obtained by adding and subtracting a suitable multiple of the combined uncertainty to and from the measured value, or in other ways, for example, using the methods described in the Supplement 1 to the GUM [24].

The GUM (6.2.2) goes to some lengths to distinguish such intervals from conventional ‘confidence intervals’ as introduced in [25] in the context of frequentist statistics [Neyman’s seminal paper defining ‘confidence intervals’ [25] in fact limits the analysis to two cases: ‘(ia) The statistician is concerned with a population, π , which for some reason or other cannot be studied exhaustively. It is only possible to draw a sample from this population which may be studied in detail and used to form an opinion as to the values of certain constants describing the properties of the population, π ,.... (ib) Alternatively, the statistician may be concerned with certain experiments which, if repeated under apparently identical conditions, yield varying results’. GUM is concerned with a much broader range of conditions, including experiments which cannot be repeated under identical conditions, as in biometrics.]. The expression ‘confidence interval’ also is used in Bayesian statistics [26], but its interpretation, of course, differs in the two settings. The GUM also creates additional opportunity for confusion when (in C.2.30) it defines ‘statistical

coverage interval' as an 'an interval for which it can be stated with a given level of confidence that it contains at least a specified proportion of the population' – indeed, this sounds more like the definition of a tolerance interval than a definition of any other type of probabilistic interval [27].

Gleser provides an insightful, critical analysis of the meanings implicit in the GUM, and of its statistical underpinnings [28]. In particular, he notes that 'the ISO recommendation has been of concern to many statisticians because it appears to combine frequentist performance measures and indices of subjective distributions in a way that neither frequentists nor Bayesians can fully endorse'. The revision of the GUM, underway in Working Group 1 of the Joint Committee for Guides in Metrology (Bureau International des Poids et Mesures, Sèvres, France), hopefully may remove such ambiguities whereas continuing to provide guidance that is widely applicable and is responsive to the needs of all fields concerned with measurement, including biometrics.

6 Measurement in biometrics: its meaning, statistical control and uncertainty

In biometric technical testing, it is standard [29] to report performance in terms of 'error rates' in the recognition process – those errors being either false positive (an incorrect decision that a biometric sample and the biometric reference are from the same characteristic of the same individual when they are not) or false negative (an incorrect decision that a biometric sample and a biometric reference are not from the same characteristic of the same individual when they in fact are). False positives are generally referred to as 'false matches'. False negatives are generally referred to as 'false non-matches'. Additional interrelated performance measures include 'failure to enrol', 'failure to acquire' and throughput rates. Any of these five measurands can be set as independent variables, affecting all of the other measurements in a test. The concepts in this paper will be applied to assessment of the probability of 'false matches' and 'false non-matches', but could be easily extended to assessments of these other variables.

The fact that our measurands are 'error rates' raises linguistic difficulties as common literature on measurement uncertainty uses the term 'error' to indicate the deviation of the measurand from its 'true' value [VIM (2.16) defines 'measurement error' as 'measured quantity value minus a reference quantity value'. [1]]. In this paper we will use 'error rate' with the meaning that it has in biometric testing.

In fully automated systems, decisions are made on the basis of metrics (known as 'comparison scores') developed through the comparison of a 'query' sample [ISO/IEC 2382-37:2011 also calls this a 'probe' sample.] to a previously developed reference. We will assume [Of course, we add to this sentence the obligatory 'without loss of generality'. In the case of similarity scores, where higher values indicate closer relationship between query and reference, the direction of the inequalities in expressions (1)–(4) will change.] that the comparison scores are 'distance' metrics in the sense that query samples and references which are 'mated' (i.e. from the same characteristic of the same individual) will have smaller comparison scores than query samples and references that are 'non-mated' (i.e. from different sources). For both mated and non-mated comparisons, testing develops the statistical distribution scores, s , such that for any specific observed value, S , we

will know the cumulative probabilities

$$\Pr(s > S | \text{mated}) \quad (1)$$

$$\Pr(s \leq S | \text{non-mated}) \quad (2)$$

over all observed values of S when the truth state ('mated' or 'non-mated') is known.

Under the simplest decision criterion, a threshold, τ , is selected, such observed values of S which are equal to or below τ are labelled as mated comparisons and values above τ are labelled as non-mated. The threshold, τ , is chosen such that the pair of values

$$\Pr(S > \tau | \text{mated}) \quad (3)$$

$$\Pr(S \leq \tau | \text{non-mated}) \quad (4)$$

is deemed to be both operationally and politically acceptable. The value in (3) represents the probability of wrongly labelling as 'non-mated' comparisons that are actually mated and is called the 'false non-match rate' (FNMR). The value of (4) represents the probability of labelling as 'mated' comparisons that are actually non-mated and is called the 'FMR'. These probabilities depend upon accurate characterisation of the distributions in (1) and (2). In characterising these two error probabilities in a biometric system, the values of (3) can be plotted against the values of (4) as τ increases to produce traditional receiver operating characteristic (ROC) curves [The traditional ROC plots the complement of the FNMR (the 'hit rate') against the FMR (called the 'false alarm rate').] or more recent detection error tradeoff (DET) curves [30].

The distributions of (1) and (2) can also be used to create other decision criteria for any observed value, S , such as 'likelihood ratios' [31] and 'minimum discrimination information' or 'divergence measures' [32, 33]. Distributions of (1) can also be normalised against distributions of (2) and vice versa using a variety of techniques [34]. Regardless of the decision criteria employed, the fundamental measures of interest remain the probabilities as expressed in (1) and (2). Uncertainty owing to lack of control over any of the influence variables affecting (1) and (2) will be reflected in uncertainty in the repeatability and reproducibility of the observed frequency of decision errors regardless of the decision criterion.

Both (1) and (2) are conditional probabilities based on knowledge of the ground truth: whether the comparison is actually mated or non-mated. Of course, in operation, we would like the inverse probabilities – that is the probability that the comparison is mated or non-mated given the comparison score observed. This inversion requires knowledge of the prior probabilities of mated and non-mated comparison, as well as the distributions in (1) and (2). Estimation of the application-dependent priors in biometric systems has been highly controversial, particularly when decisions depend upon a search strategy [35–37]. Discussion of their associated uncertainty is outside of the scope of this paper, but any attempt to answer the question, 'How confident are we regarding the truth of the reported result of this biometric comparison?' would have to consider the uncertainties from all of these sources (including the priors) and is beyond the level of current understanding for any documented biometric system.

Predictive estimation of the numbers of false positive or false negative decisions expected by a biometric system in

some time interval requires not only estimation of the throughput and priors, but also a system model expressing how thresholds and priors will be used with (1) and (2) to create system decisions, which might involve multiple comparisons using correlated query samples and multiple references [38]. These models will be dependent upon whether the system is designed for ‘verification’, ‘identification’ or some other application of automated recognition. Uncertainty in the model and throughput rate, as well as the priors and distributions, will contribute to uncertainty in estimates of numbers of false positives and false negative decisions in the time interval. The remainder of this paper is concerned only with uncertainty in the estimation of (1) and (2) regardless of how those statistics are used to make comparison decisions and regardless of the system employing them or the operational prior probabilities.

7 Statistical control and uncertainty in technology tests

Philips *et al.* [39] developed the concept of three levels of testing in biometrics: technology, scenario and operational. This concept made its way quickly into the literature and standards in the field. The next sections consider uncertainty in technology, scenario and operational testing sequentially.

A technology test uses a pre-collected database of biometric samples and reports the ROC or DET for each software package against that database [We are specifically avoiding the use of the word ‘algorithm’ because a software package for biometric recognition will contain many different algorithms – for example, algorithms for signal detection, segmentation, feature extraction, model creation and comparison. Consequently, our technology test results will not tell us which of the algorithms in the package are functioning well and which are defective, but rather will reflect the function of the software package as a whole, to echo Duhem [9].]. Hence each software package is tested against exactly the same data and our results tell us about the efficacy of the software with respect to the database. Examples include most of the NIST biometric test programs, such as the Speaker Recognition Evaluation, the Multi-Biometric Grand Challenge, the Proprietary Fingerprint Template Testing and Minutiae Exchange Test programs.

If the software is deterministic, and there is a prescribed way to use the test data, then barring software or hardware limitations that may bias the results, and assuming pristine bookkeeping by the testing agency, the results of a technical test would be completely repeatable [Our experience in conducting tests has demonstrated the fallacy of these assumptions: memory misallocations in software often cause code to produce different results across different machines or even across different runs on the same machine, and bookkeeping is not always flawless.]. Running a software package multiple times against the same database would, under these assumptions, result in the same performance measures. We can state number of false match and false non-match errors (or rate of errors per non-mated and mated comparisons, respectively) for each package when run on each database.

As an example, Table 1, from the Proprietary Fingerprint Template Evaluation Program [40], shows results for true accept rate ($TAR = 1 - FNMR$, where $FNMR$ is the false

Table 1 From [40] ‘TAR at FAR = 0.0001’

	DHS2	DOS	POE	POEBVA
D	0.9917	0.9845	0.9955	0.9932
F	0.9893	0.9944	0.9979	0.9979
H	0.9870	0.9978	0.9993	0.9994
I	0.9904	0.9978	0.9992	0.9992

[From the February 26, 2010 update at http://fingerprint.nist.gov/PFT/tables2f_121109.pdf]

[Here, use of the term ‘FAR’ (false acceptance rate) in [40] is to be interpreted as the FMR.]

non-match probability given by (3)) at a threshold producing a FMR given by (4) of 0.0001 for four packages (D, F, H and I) against each of four databases (DHS2, DOS, POE and POEBVA) collected under different conditions [NIST generally does not report results in ISO/IEC compliant formats.].

Under the assumptions in the previous paragraph and allowing no changes to either the software or the database, these results would be completely repeatable within the same lab and reproducible across labs [However, for reasons of data protection of the personal biometric data and to maintain the data as sequestered for testing purposes only, it is generally not possible to share databases across laboratories. An equivalent ‘reference’ database of the same difficulty cannot be constructed by a second test lab, since the conditions under which the original data were collected cannot be replicated.]. Table 1 shows that changes in either software or database result in unpredictable changes in reported results [For example, Software Package D performs better on database DHS2 than on DOS, whereas Software Package F performs worse. Therefore we cannot predict even the direction of the influence on the performance measurements when making changes to algorithms or databases.].

Even if completely repeatable and reproducible, these results will contain uncertainty in the statement of the measurand (comparison errors) resulting from the comparison process. The comparison process takes place in the context of a matching key (the assumed ‘ground truth’) – the answer sheet holding the truth values. The measured quantities reported in the test – the comparison errors as discrepancies between the algorithm decisions and the matching key – are really proxies for the stated measurands, which are the ‘error rates’ of the algorithm against the database. An assumed equivalence between the comparison quantities measured and the stated measurands would require complete accuracy and statistical control of the matching key for the database. NIST has estimated the labelling errors in government databases to be 1 in 1000 [41]. It is quite common in the course of all biometric evaluations for the matching keys to be updated as mistakes in the original are suspected, then confirmed through analysis methods independent of the recognition software.

GUM is not willing to give guidance on the issue of labelling errors, stating

Blunders in recording or analysing data can introduce a significant unknown error in the result of a measurement. Large blunders can usually be identified by a proper review of the data; small ones could be masked by, or even appear as, random variations. Measures of uncertainty are not intended to account for such mistakes [22].

The GUM's general probabilistic approach to measurement uncertainty, however, can be extended to include the consequences of labelling errors. Indeed, making either the most simplistic assumption (that labelling errors occur independently and with constant probability throughout all the items in a database), or assumptions that may more realistically capture the patterns of interdependence to be expected (illustrated below), statistical models can be built that produce assessments of uncertainty attributable to this particular cause, even if 'by construction' the mislabellings are systematic, not random.

Also, considering labelling errors as one contribution to measurement uncertainty is consistent with the approach used in [41]. That approach was to refuse to report results below the measurand uncertainty level introduced by the suspected level of labelling errors. Figures in [41] intentionally cut off all graphs at measured values below 0.001 for either type of error, allowing the data line to disappear off the boundaries as the labelling error rate was given a subjective probability by the experimenters of exceeding the value of the measurand.

So the 'unit of empirical significance' for a technology test contains the software package, the database and the matching key along with all the other factors in our 'web of belief' [Of course, each of these areas is a 'unit of empirical significance' in itself, containing massive requirements for statistical controls. The oft-told story of the 1987 KING speech database collection illustrates how lack of statistical control in database collection can disrupt an entire government technical test program. About halfway through the KING collection, a minor equipment change was made that altered the spectral content of the speech recordings, creating a divide in the data that disrupted data analysis for years until the problem was understood. See Godfrey J, Graff G, Martin A. Public databases for speaker recognition and verification. ESCA Workshop on Automatic Speaker Recognition, Identification and Verification; 1994.]. If the only source of uncertainty is the labelling mistakes in the key, how can the impact on performance results be quantitatively assessed? Often there is enough redundant information in a data collection for the experimental team to eliminate most labelling mistakes, but ultimately we can never know what residual mistakes remain and will be required to make a Type B assessment.

These labelling mistakes will be distributed in some unknown way over both incorrect mating and non-mating information. Incorrect non-mating information (to say two samples are from different sources when they are from the same source) will cause an accurate software package to appear to have a higher number of false matches. We'll call this 'Type 1' mislabelling. Incorrect mating information (to say two samples are from the same source when they are not) will lead an accurate software package to appear to have a higher number of false non-match errors. We will call this 'Type 2' mislabelling.

The statistical impact of the two types of labelling mistakes will be different. Consider a database with n samples, of which there are actually m mated pairs. Suppose the key wrongly considers k mated pairs as not mated, and wrongly considers j pairs as mated when they are not. The key will consider $m+j-k$ pairs to be mated. A perfect software package will fail to match the j pairs and be assessed a FNMR of $j/(m+j-k)$. It will match the k pairs and be assessed a FMR of $k/[n(n-1)/2 - (m+j-k)]$. An estimate of one labelling mistake per 1000 samples implies that $(j+k)/n = 0.001$, but there is no reason to suppose any

partitioning of these mislabels over Types 1 and 2, so there is insufficient information to even guess at the quantitative bias introduced into FNMR and FMR except to say that reported values will be biased on the high side.

These mislabellings in the key will not be random, but rather systematic – for example, Type 2 mislabelling of twins as the same person or Type 1 mislabelling of different fingerprints from the same person as the same fingerprint. Therefore the stronger (more accurate) software packages will be more likely to deliver results conflicting with the mistaken key, whereas weaker packages may agree with the key. Hence, the impact of the labelling mistakes on measured performance error rates will not be uniform across all software packages and will force any meaningful coverage interval to expand primarily in one direction. The 'true' value of measurands (i.e. the comparison error rates) are likely to be smaller than measured. Furthermore, uncertainty in the reporting of the FMR may require a change in the threshold at which the FNMR is reported. An FMR reported higher than the correct value because of labelling mistakes would further push the reported value of FNMR to be higher than the correct value at the stated FMR.

Returning to Table 1, there may be the temptation to make broader statements about the results, saying the Software Package D averaged a TAR ($1 - \text{FNMR}$) of 0.9912 over the interval of (0.9845, 0.9955) at a measured FMR of 0.0001, thus indicating some uncertainty owing to databases, but not to matching keys. Again, this uncertainty is asymmetric and cannot be expressed as $a \pm b$ for any software package. The GUM's prescription of intervals of this form, however, cannot be divorced from the underlying assumptions that validate such prescription, one of whose central tenets is that the measured value is like an outcome of a Student's t random variable (possibly, but not necessarily, with a large number of degrees of freedom). The Supplement 1 to the GUM provides the means to tackle situations where such assumption of symmetry is not applicable [24].

Further, because the size of the various databases is not known, such an average may overemphasise results from the smaller databases in the ensemble. The mean value might be wrongly taken to be the TAR which would be found if the databases were consolidated into a single test set. Consequently, calculation and reporting of the variance of the measures around the mean across all databases would be of no value. We must not be tempted to implicitly extend these results to include any possible databases. The uncertainty reported as above applies only to the tested databases under the assumption of no labelling errors, which we have shown introduces systematic, not random, variation in our measurements. In short, it would be highly misleading and scientifically wrong to make the overly generalised statement 'Software Package D has a TAR of 0.9912 at FAR = 0.0001'. A more correct statement would be 'Software Package D had a measured TAR at FAR = 0.0001 of 0.9845 to 0.9955 across the various databases tested. Labelling errors could cause the true TAR to be as much as XX higher than measured' (where XX is the estimate of the experimenter using both Type A and Type B analysis, including the experimenter's experience and general knowledge).

Of course, the above paragraph could be repeated for packages D, F, H and I for any one of the databases to give, for instance, a measure of database difficulty. However, trying to make a general statement for all tested software packages across all tested databases would

compound the problems already stated. A technology test cannot be used to give performance metrics for a general technology (say automatic fingerprint recognition) independent of reference to a specific test database. Therefore technology testing on artificial or simulated databases tells us only about the performance of a software package on that data. There is nothing in a technology test that can validate the simulated data as a proxy for the 'real world', beyond a comparison to the real world data actually available. In other words, technology testing on simulated data cannot logically serve as a proxy for software performance over large, unseen, operational datasets.

However, what about random uncertainty owing to small database size in a technology test? Such uncertainty stems from the random selection of a subset of data from a much larger database. The uncertainty then includes a component that is attributable to the corresponding sampling: if this should be random, then it is the object of Neyman's defining work [25]. One approach used in the biometrics literature [42] is to employ a 'divide-and-conquer' strategy whereby a single database is randomly partitioned into subsets, and the variability of the performance (error rates) of the matching software among the different subsets be used as intimation of its operational acumen. This is similar to the idea of statistical 'cross-validation', where models are developed based on one part of the data, another part being set aside, and their performance is assessed using the portion set aside. However, in technology testing, test data is generally not a randomly selected subset of a larger database. Rather, if a larger database of biometric samples does exist, a selection of the test database is made to systematically remove problems, such as 'poor quality' or blank images [The NIST Iris Comparison Evaluation is an example of a test set created by removing problem images selectively from a larger dataset. See Appendix A-1.1 of Phillips, *et al.*, 'FRVT 2006 and ICE 2006: Large-scale Results' NISTIR 7408, March, 2007.]. Some of the issues of dataset selection are well considered in Appendix E of [41]. So lacking a larger database from which the test dataset was randomly selected, there are no associated random influences connected with the measures reported for the test database.

Of course, there could be the hypothetical extension of a database that might have been, but was not, collected. This would require the untenable assumption that 'unit of empirical significance' would remain completely unchanged except for size and that all of the statistical controls on the smaller collection would be applied to the larger collection with no problems of scaling. Experience teaches, however, that to acquire a dataset of 30 000 samples requires a broader search for very different sources than to opportunistically acquire 300 samples. Therefore the 'hypothetical extension' argument does not create any larger database within the same unit of empirical significance as the data actually acquired, meaning that there is no established need for Type A random uncertainty calculation for technology tests. Values extrapolated from our test database to some hypothetically identical but larger database under the assumption of non-existing statistical controls would measure 'nothing at all'. Consequently, questions of the type commonly stemming from Table 1, 'Are the measured differences in performance of the tested algorithms statistically significant given the size of the test?' is ill-posed, although the question 'Are the measured differences significant given the expected occurrence of labelling errors?' will have merit requiring Type B analysis.

Is there any hope of inductively extending the results of a technical test more broadly to any other algorithms or databases? A Type B systematic uncertainty evaluation after consideration of changes in the unit of empirical significance and statistical controls over its tangible elements might be of value, provided that the specifics of the changes could be given, but such a 'guesstimate' should not be sanctified in an emperor's cloak of imagined analytic rigor.

We conclude this section by stating that a technology test reports the measurands (generally the values of (3) and (4)) of a specific software package in relationship to a labelling key for a specific database of biometric samples. Any attempt to use these results as a predictor of future behaviour of any biometric system must be done with full appreciation of the limitations in these values owing to the uncertainties of the tests.

8 Statistical control and uncertainty in scenario tests

A 'scenario test' moves us from the use of previously collected data to the process of data collection from human subjects. The introduction of human data subjects directly into the test environment causes great changes in the unit of empirical significance and makes scenario testing a social science – as much about the data subjects as about the technologies.

Scenario evaluations measure overall system performance for a prototype scenario that models an application domain... Scenario evaluations test complete biometric systems under conditions that model real-world applications. Because each system has its own data acquisition sensor, each system is tested with slightly different data. One scenario evaluation objective is to test combinations of sensors and algorithms [39].

In a scenario test, the experimenter controls to some extent the conditions of the collection and the test population, thereby exerting some influence on the statistical controls of the tangible elements within the unit of empirical significance. The goal is to model an application system in a similar environment and with a similar population, but what conditions should be included in the model?

The measurands, (3) and (4), are found by observing the numbers of false negative and false positive decisions at a set test threshold during two tests – one involving only mated comparisons and one involving only non-mated comparisons – then averaging those numbers over the total number of mated and non-mated transactions, respectively. The transactions often involve multiple correlated query samples and multiple reference samples, or a reference created from multiple samples selected after a quality assessment, so the calculation of values of (3) and (4) are generally not commensurate with calculations of (1) and (2) in a 'technology' test involving only the comparison of single query samples to single references without use of a threshold in the collection process. Influence quantities [An Influence quantity is a 'quantity that is not the measurand but that affects the result of the measurement' [1].] extend to number of samples and sampling conditions, persons sampled, time of sampling and the affect of the test threshold upon subject behaviours. We lack both metrics for assessing the expected variability of these quantities

between tests and models for converting that variability to uncertainty in measurands.

A recent NIST handbook [43] suggests that factors attributable to or affecting the data subjects which influence technical test measurements include, but are not limited to: data subject age, gender, experience and height; the instructional materials presented the data subject; the placement, height and angle of the collection device; the temperature, humidity, lighting and noise of the collection location. For a scenario test, the values of these factors in the operational environment to be modelled may not be known or controllable by the experimenter.

In addition to this list of general, technology-independent, but human-related factors to be controlled, each specific recognition technology (iris, face, voice, fingerprint, hand etc.) will have specific factors that must be within a state of statistical control. This list of factors is not well understood, although ample work in this area is continuing.

For example, recent analysis of iris [44] and face [45] recognition test results shows us that to report false match and false non-match performance metrics for such systems without reporting on the percentage of data subjects wearing contact lenses, the period of time between collection of the compared image sets, the commercial systems used in the collection process, pupil dilation and lighting direction is to report 'nothing at all' [However, this is not to say that our listing of relevant sources of uncertainty in this list is complete. So even if we report and control these factors, we may be still unable to repeat or reproduce these results and may indeed have measured nothing at all.]. Our reported measurements cannot be expected to be repeatable or reproducible without knowledge and control of these factors, most of which cannot be enumerated or quantified at our current level of understanding.

However, beyond the observable and easily quantifiable aspects of the collection conditions, such as lighting and acoustic noise, and of the data subjects, such as height, handedness and presence of glasses, the less tangible influence quantities of human attitudes, perceptions and behaviours of both the data subjects and supervising experimenters affect the measurands and thus the repeatability and reproducibility of scenario test results. The story of the 'lost' 1993 scenario test from Sandia National Laboratory (SNL) is illustrative. In the widely-referenced 1991 scenario test of several biometric access control devices [46], SNL determined the comparison error rates for hand geometry. At a threshold of 100 vendor-proprietary units, the device achieved an 'equal error rate' of 0.2% for both measurands. ['Equal error' or 'cross-over' error rates allow us to report a single value for the two measurands, but are generally not operationally relevant. A decision threshold producing the 'equal error rate' would only be rational if the product of the costs and priors of a false rejection were equal to the product of the costs and priors of a false acceptance, which is an extremely unlikely condition. Comparing tests or devices in terms of 'equal error rate', which is a convenient if operationally questionable measure, introduces confusion into our community.] In the 1993 test [47], the 'equal error rate' of 5% was achieved at the same threshold of 100 [That the equal error rates in the two tests occurred at the same threshold makes these results comparable.]. The manufacturer asked for an explanation of the extreme increase in the measurand in the second test, given that the devices tested were nearly identical. An internal enquiry at SNL determined that one data subject had been conducting

undocumented 'experiments' with hand placement. So the test design was such that the behaviour of one data subject was able to increase both measurands by a factor of 25 and prevent repeatability of the first test. The second test report was withdrawn and no copy was available within SNL until a 'preliminary draft' was discovered in the archives of the National Physical Laboratory and a copy returned to SNL. The story illustrates that the uncertainty in our measurements in a scenario test extends to the attitudes and behaviours of the data subject.

In 2000, the National Physical Laboratory [6] tested a very similar hand geometry device from the same manufacturer, obtaining measurands of 0.7% (FMR) and 0.5% (FNMR) at the threshold for (3) and (4) of 100 vendor-proprietary units. These results show that the test repeatability and reproducibility observed in technology tests are lost in scenario testing because of the loss of statistical control over a wide range of influence quantities.

An additional complication in a scenario test is that the software package may require modifications such that the comparison scores can be fully observed. This means that the software package being tested is itself only a proxy for the package that would be installed in the target operational environment with additional operational controls and interfaces. As in technology tests, complete statistical control over 'ground truth' – accurate knowledge of which comparisons are mated and which are non-mated – may be difficult to attain.

In addition, scenario testing will usually take place with a decision threshold in place that will supply feedback to both the experimenters and the data subjects, both of whom will generally adjust their behaviour to produce favourable outcomes at the test threshold. Devices may also continue to collect samples from the data subject until a comparison score reaches the threshold. This latter condition would mean that 'off-line' and 'on-line' tests to determine a score distribution for non-mated comparisons would lead to different results and both would be dependent on the thresholds used in the scenario test. This implies that 'impostor' testing should be done as part of the scenario, not with 'off-line' testing with data collected during the test. Even so, test results shown on ROC or DET plots do not represent well the measurands at thresholds other than that used for the test. Consequently, scenario tests should be done at decision thresholds realistic to the target operational environment. Such thresholds are set in operation after a subjective analysis of the risk implied by each kind of error. Consequently, one important influence quantity in our unit of empirical significance will be collection threshold that reflects the anticipated level of risk tolerance by the system operators. Additional influence quantities affecting scenario test measurands are given in [29, 48]. Our inability to apply concepts of statistical control to any or all of these factors will increase the level of uncertainty in our results and translate to loss of both repeatability and reproducibility.

Therefore if the purpose of the scenario test was in fact to predict performance in the modelled operational system, we should accept that there is a high level of uncertainty in the measures. At the current level of understanding of the influence factors in both scenario and operational environments (owing to lack of both experience and data), the focus should be to develop Type B uncertainty assessments. Test data from scenario evaluations should not be used as input to mathematical models of operational environments that require high levels of certainty for validity.

9 Statistical control and uncertainty in operational tests

An operational test seeks to determine technical performance metrics in a real, application environment. The problems with such testing are legion and few operational tests have been performed. Of those performed, only a very few have been published [7, 8, 49], owing to the sensitivity of system management to what are perceived as security-related metrics. The lack of extensibility of technical and scenario test measurements to operational environments has been noted for as long as there has been third-party testing in biometrics [50].

Moving from a laboratory to an operational environment raises issues common to other scientific endeavours, particularly those in the psychological and sociological communities [51–54]. The general criticism of laboratory testing in these fields, as in biometrics, is that ‘human behaviour outside the laboratory cannot be predicted from what has been learned in the laboratory’ [51]. Our experience with all forms of biometric tests is that our experimental findings are limited to the conditions of a specific study, which is contrary to the general goal of science to produce results generalisable to both larger populations and to populations and conditions beyond those studied. Lucas [52] argues that results of specific experiments can only be extended through the application of a theoretical model for doing so [Lucas [52] argues that extending results from test subjects to larger populations is not simply a matter of careful random sampling of the larger population but also requires a model of how results vary with population heterogeneity. Mook [53] argues that we can appropriately test and develop theories without ‘random’ sampling and without defining a target population because extensibility to larger groups is not our only goal in testing. The fact that a particular result can be obtained from some members of the population may teach us about influence variables.]. Thus, he argues that the limitations of test results to the conditions of specific tests are not the fault of the test methodologies, but rather owe to the lack of an extending model. Such a model would need to consider all of the influence variables, most of which remain unknown for any biometric technology, application environment and target population. What is gained from laboratory testing (technology and scenario) is an understanding of components of the operational processes including the influence variables, making such tests valuable even in the absence of extensibility of results to operational environments [53]. The controlled environment of the laboratory, although differing radically from any operational environment, may be the only way in which such variables can be studied, even if we lack an adequate understanding of all of the variables that must be controlled [54].

Lacking such extending models, biometric testing groups hold no illusions that the results of technical or scenario tests can be extended across populations and test conditions to predict operational performance without a major increase in the uncertainty of the measurands of interest.

Operational testing inherits all of the uncertainty of technology and scenario tests, but with added problems of statistical control over all population and environmental elements in the unit of empirical significance. Of particular difficulty in operational testing is determining the ‘ground truth’ of any comparison. In an operational environment, the data subject will make some claim (perhaps only

implicitly) as to being or not being the source of a biometric characteristic in the database [For linguistic consistency, the ‘claim’ is always from the point of view of the data subject. For example, in a watchlist environment, a data subject may not realise that a biometric system is in operation. However, if presence on the watchlist is pejorative, we will say that the data subject makes an implicit claim not to be the source of any biometric characteristic in the database.]. We can assume that an acceptance of this claim, regardless of whether true or false, by an operational data subject will go unreported. When a software decision results in ‘rejection’ of the claim of an operational data subject, the data subject may be informed of the decision by some applications, allowing redress. Under the questionable assumption that the redress process always can always correctly classify the original claim based on the original presentation as ‘true’ or ‘false’, the ‘system false rejection’ rate can be estimated. Of course, this assumption is incorrect, leading to measurement uncertainty.

However, those with an ‘accepted’ claim, whether that claim is true or false, will not seek redress. Therefore in an operational system, we will have a way of estimating the ‘system false rejection’ rate, but not the ‘system false acceptance’ rate. Consequently, most tests of system false acceptance involve the staging of transactions with the presentation of false claims. Such staging may lead operational tests of system false acceptance to resemble scenario tests conducted on the operational system.

However, there is another hidden problem here. False ‘acceptance’ or ‘rejection’ refer to a claim, perhaps by the data subject or by another actor in the transaction, as to whether the comparison of query and reference are mated or non-mated. For example, in a ‘verification’ system, the data subject makes a positive claim that the comparison made will be mated – that is that the query samples presented will match a reference which is usually specifically identified or supplied. In other applications, such as ‘identification’ applications intended to prevent creation of multiple references for a single individual, the data subject will make a negative claim that the comparisons made will not be mated – that is a claim that the query samples presented will not match any reference to which it is compared. Depending upon search strategy implemented by the system software, decisions in either verification or identification systems may involve multiple comparisons of multiple query samples against some number of references in the database. So mapping ‘rejection’ and ‘acceptance’ to ‘mated’ and ‘non-mated’ comparisons for estimation of (1) and (2), and the false non-match and FMRs of (3) and (4) at the operational threshold, will require knowledge of the specific search strategy implemented in the software, including the number of both mated and non-mated comparisons made for each search. Such knowledge is generally unavailable to the test agency, so the only directly observable metric is the rate of rejected claims (either positive or negative) done in the absence of total certainty over ground truth.

In a scenario test, we seek to control the presentation of the biometric characteristic by the data subject. This control can be through training or supervision or even the exclusion of data deemed not given in ‘good faith’ [See [29] for discussion on excluding data based on lack of ‘good faith’ by the test subject.]. Measurands are based on data acquired and even edited according to such rules. In an operational test, such presentations cannot be actively controlled, nor

can judgments be consistently made on the 'good faith' of the data subjects. In the case of 'genuine' transactions (i.e. those with a correct claim) in an access control application, there is not clear way of distinguishing poor faith attempts from confusion over the proper use of the equipment or simply lack of motivation. Indeed in some cases, for example when the biometric characteristics are mostly obscured, rejecting the claim would be the appropriate action for the biometric recognition software, and so ought not be considered a 'false' rejection. Moreover, the rejection might be for reasons not related to the biometric comparison, such as for expired credentials, which may not be observable by the testing agency. So the reported measurand is really the 'system rejection rate'. Rather or not that rejection should be considered as 'false' depends upon system policies, data subject behaviours and non-biometric considerations [8, 49] that are generally not observable to the experimenter. The rate of system rejections determined to be 'false' will depend upon how 'false' is defined in relationship to expected adherence to policies of the system. Variation in definitions and system policies will add to uncertainty in reproducibility and repeatability of measurands.

We can conclude that the three types of tests (technology, scenario and operational) are measuring incommensurate quantities and therefore should not be at all surprised when the values for the same technologies vary widely and unpredictably over tests of different types.

10 Summary

If automated human recognition, or 'biometrics' in our sense of the term, is to be considered as a science, then it must be subjected to the same requirements in testing and reporting as any other discipline within the scientific community. The international statistical and metrology communities have focused a century of effort into explicating scientific testing and reporting: what measurements are desired, what measurements can be made in a test, what measurements are actually made, how these various measurements are related and how the attendant uncertainties can be characterised and quantified. The International Organisation for Standardisation document, 'GUM', was created to supply a framework for the consideration of measurement practices and the evaluation of uncertainties resulting from laboratory and field measurements. This document is as relevant to biometrics as to any other scientific field.

Technical testing in biometrics has historically focused on recognition error rates of two types: false positives (also called false matches – an incorrect decision that two biometric samples are from the same characteristic of the same individual when they are not) and false negatives (also called false non-matches – an incorrect decision that two biometric samples are not from the same characteristic of the same individual when they in fact are). These measurements have historically proved to be neither exactly reproducible nor repeatable except in very limited cases of repeated execution of the same software package against a static database on the same equipment. Further, the observed measurands in technology, scenario and operational tests are quite different. Accordingly, 'technology' test metrics have not aligned well with 'scenario' test metrics, which have in turn failed to adequately predict field performance of operational systems.

All test results carry uncertainties related to measurand definitional incompleteness and systematic variation that have

not been addressed by our biometrics community. This paper has applied the concepts developed by the statistical and metrology communities, as articulated in GUM, to explain why this is to be expected and how uncertainty assessment must be extended beyond 'confidence intervals' – the computational bounding of random variation owing to small test size.

We have shown that technology tests only measure proxies for comparison error rates, how scenario tests can introduce population and environmental variations, and how operational tests measure an overall system performance incommensurate with the measurements in the other forms of testing.

Specifically, our primary remarks can be summarised as:

1. 'Uncertainty', a broader concept than 'error', is doubt about how well a test result represents the quantity it is said to measure. Uncertainty can exist even in the absence of error, in the sense of 'mistake' and extends beyond the analytic calculation of expected random variation caused by limited test size.
2. A central source of uncertainty in a measured value, and in its repeatability and reproducibility, is definitional incompleteness in specifying all of the factors influencing the measurement.
3. What we actually measure is usually only a proxy for what we want to measure.
4. How we control, measure and report the values in a test must reflect how we expect those values to be used by others. In other words, our testing and reporting must take into account, and state, how we expect the results to be used.

Improving our ability to understand, predict and control the performance of biometric systems in operational environments will require more through study of the human and environmental influence variables, greater attention to definitional completeness in our measurands, and more experience in estimating the systematic uncertainties introduced by changes to the software, hardware, environmental and human factors encountered in the 'real world'.

Moving forward, it would be our recommendation that research efforts in our field:

1. State clearly and as completely as possible the conditions of the test, including information on test subjects and the conditions of their recruitment, environmental variables, data collection hardware/software, number and duration of data collection sessions, and data editing and pre-processing protocols.
2. Define clearly the measurands of interest and the proxies used for those measurands (i.e. the comparison of images from different sources as a proxy for a 'zero effort impostor attempt').
3. Discuss potential sources of systematic uncertainty separating the reported measurements from those of the intended measurand.
4. Limit the scope of the conclusions to the specific system under test, the specific conditions of the test and the specific purpose for which the test was conducted, avoiding comparisons of the putative measurand under the reported test conditions to measurands in other systems under other conditions.

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